SpendLog AI

Use Case & Diagrams Report

Actor Interactions, UML-style Diagrams & Process Flows

Document No: DOC-04
Project: Agentic Expense Tracker (SpendLog AI)
Repository: https://github.com/Harish-S28/Agentic-expense-tracker
Prepared for: Project Documentation / README reference

1. Actor

SpendLog AI has a single actor: the **User** — an individual tracking personal expenses under one of five personas (Student, Doctor, Business Person, Parent, Employee). There is no separate admin role; the system is designed as a single-tenant personal tool per deployed instance.

2. Use Case Diagram

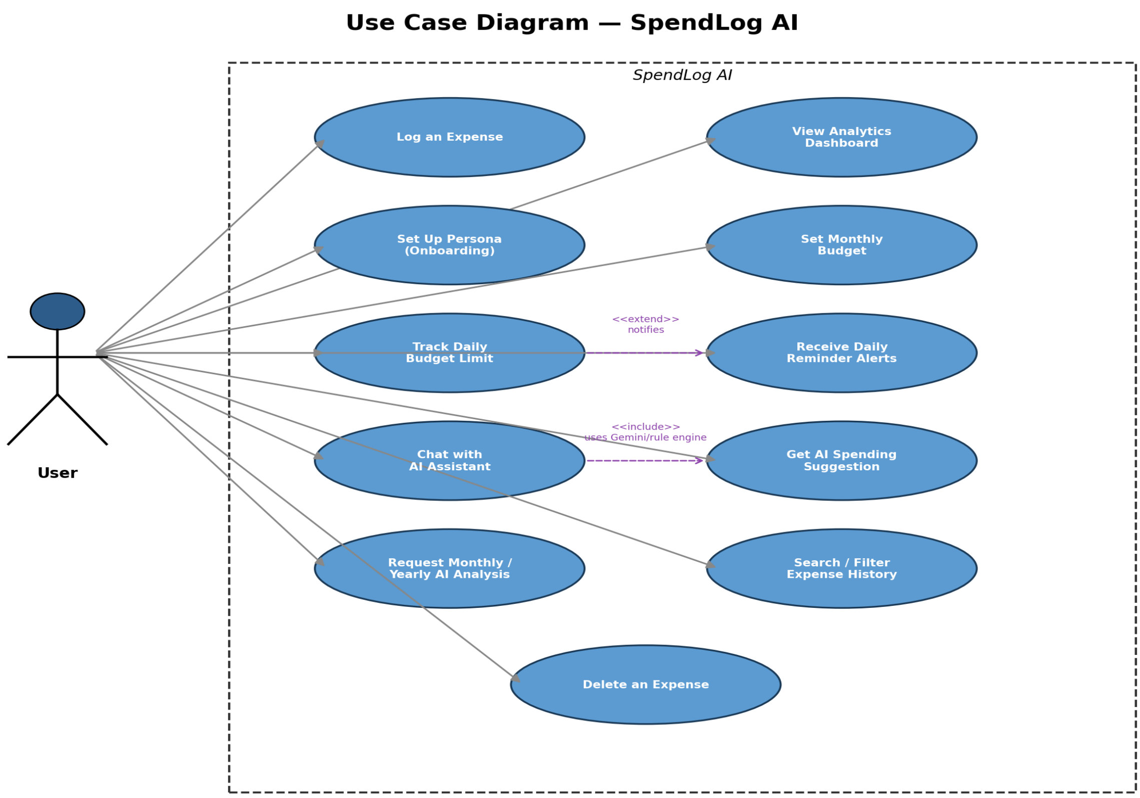

Figure 1. Use case diagram — User and the eleven primary use cases of SpendLog AI.

3. Use Case Descriptions

UC-1. Log an Expense

Actor	User
Preconditions	App is running; user is on the dashboard.
Main Flow	1. User opens the add-expense form. 2. Enters amount, category, note, date. 3. Submits. 4. System stores the row a
Postcondition	New expense persisted; AI suggestion shown as a toast.
API Route(s)	POST /api/expenses, POST /api/ai/suggest

UC-2. View Analytics Dashboard

Actor	User
Preconditions	At least one expense exists (optional — empty state supported).

Main Flow	1. User opens Analytics. 2. Optionally filters by month. 3. System returns total, category breakdown, top spending da
Postcondition	User sees an up-to-date visual/summary of their spending.
API Route(s)	GET /api/analytics

UC-3. Onboard with a Persona

Actor	User
Preconditions	First run, or user revisits profile settings.
Main Flow	1. User enters name, selects a profession/persona (Student, Doctor, Business, Parent, Employee), optionally enters
Postcondition	All future AI responses are conditioned on this persona.
API Route(s)	GET/POST /api/profile

UC-4. Set Monthly Budget

Actor	User
Preconditions	Profile may or may not be set.
Main Flow	1. User enters a target monthly budget amount. 2. System stores it keyed to the current 'YYYY-MM'.
Postcondition	A daily limit and carry-over engine become active for that month.
API Route(s)	POST /api/budget

UC-5. Track Daily Budget Limit

Actor	User
Preconditions	A monthly budget has been set for the current month.
Main Flow	1. User opens the dashboard/budget widget. 2. System computes base daily limit, walks every prior day of the month
Postcondition	User sees whether they are within or over today's effective limit, plus an AI coaching tip.
API Route(s)	GET /api/budget/status

UC-6. Receive Daily Reminder Alerts

Actor	User (passive)
Preconditions	Browser has granted Notification permission.
Main Flow	1. Client polls every 60 seconds. 2. In the 9:00-9:10 AM or 7:00-7:10 PM windows, it calls the reminder-check endpo
Postcondition	User is nudged to log an expense if they haven't already today.
API Route(s)	GET /api/reminders/check

UC-7. Chat with AI Assistant

Actor	User
Preconditions	Chat drawer is open.
Main Flow	1. User types a free-form question (e.g. 'How much have I spent this month?'). 2. System assembles a financial snap
Postcondition	User receives a contextual, conversational answer.
API Route(s)	POST /api/ai/chat

UC-8. Get AI Spending Suggestion

Actor	User (system-triggered)
Preconditions	An expense was just logged.
Main Flow	1. System gathers the expense plus this month's category/monthly totals and budget. 2. AIAgent produces a 3-4 sen
Postcondition	User receives immediate, contextual feedback on the expense just logged.
API Route(s)	POST /api/ai/suggest

UC-9. Request Monthly or Yearly AI Analysis

Actor	User
Preconditions	Sufficient expense history exists for the selected period (works even with zero data — templated defaults apply).
Main Flow	1. User opens the Analysis page and selects month or year view. 2. System aggregates the relevant period's data. 3
Postcondition	User receives a full financial review for the selected period.
API Route(s)	GET /api/ai/analysis

UC-10. Search / Filter Expense History

Actor	User
Preconditions	Expenses exist.
Main Flow	1. User applies a category filter, month filter, exact-date filter, and/or free-text search across note/category fields (any
Postcondition	User locates specific historical expenses quickly.
API Route(s)	GET /api/expenses?category=&month=&search=&date=

UC-11. Delete an Expense

Actor	User
Preconditions	The target expense exists.
Main Flow	1. User selects an expense from a list. 2. Confirms deletion. 3. System removes the row.
Postcondition	Expense is permanently removed and no longer counted in analytics or budget.
API Route(s)	DELETE /api/expenses/<id>

4. Sequence Diagram — Expense Logging & AI Feedback Loop

This diagram traces UC-1 and UC-8 together, since in the real UI an AI suggestion is fetched immediately after every successful expense insert.

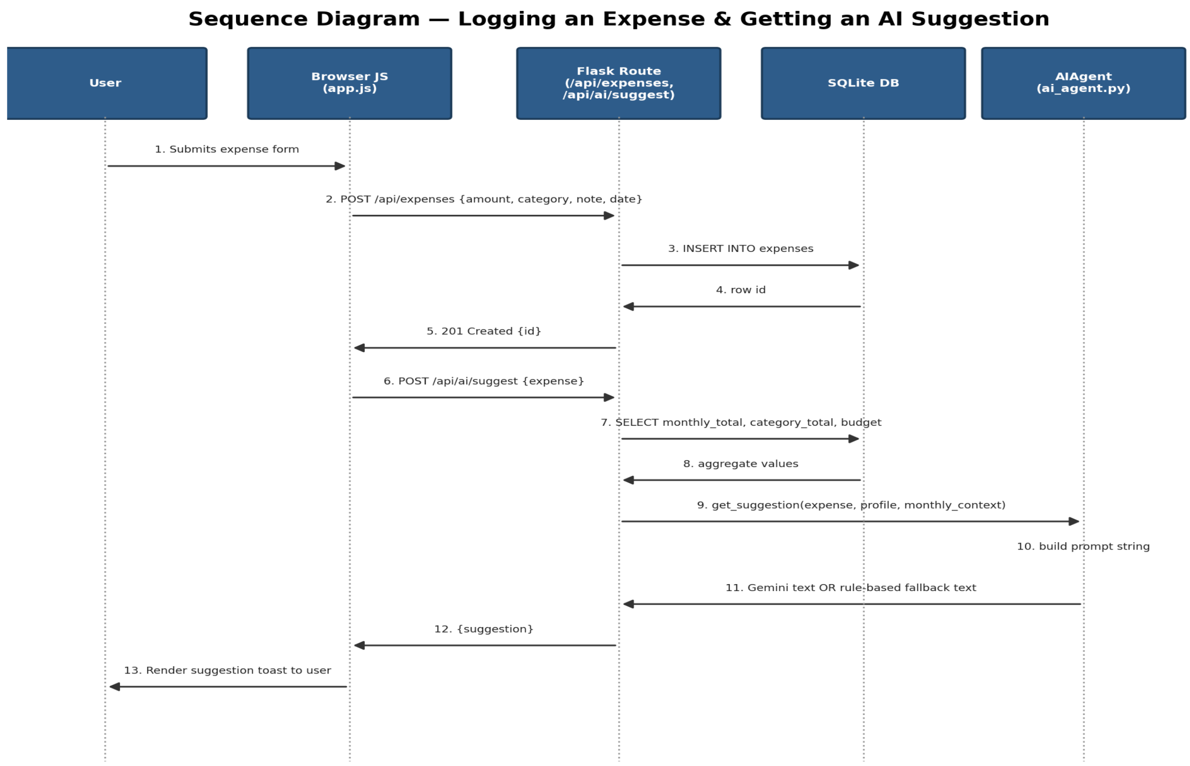

Figure 2. Sequence diagram covering UC-1 (Log an Expense) and UC-8 (Get AI Spending Suggestion).

5. Process Flow — Budget Carry-Over (UC-5)

The daily budget limit is not static: any day's under-spend becomes a bonus added to future days, and any overspend becomes a penalty subtracted from future days. The flowchart below documents the exact algorithm implemented in **_get_carry_over()** and **/api/budget/status**.

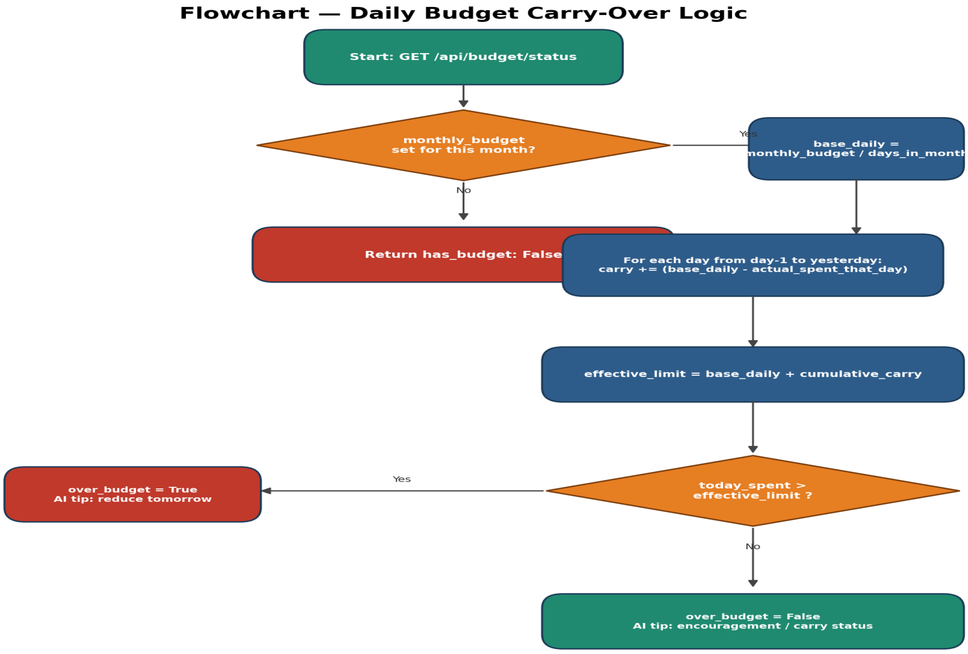

Figure 3. Flowchart of the daily budget carry-over calculation underlying UC-5.

6. Data Model Reference

All use cases ultimately read from or write to the three tables shown below; it is included here as a quick cross-reference between use cases and persisted data.

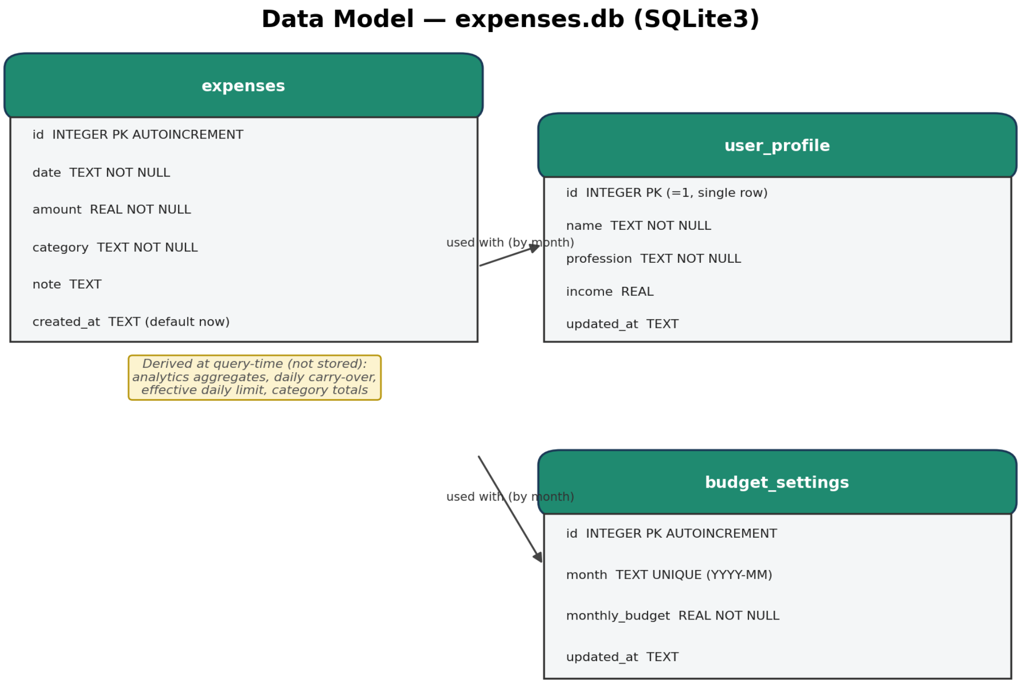

Figure 4. Data model referenced by the use cases in this report.

7. Use Case Traceability Matrix

Use Case	Primary Table(s)	AI-Assisted?
UC-1 Log an Expense	expenses	Yes (suggestion)
UC-2 View Analytics	expenses	No
UC-3 Onboard Persona	user_profile	No
UC-4 Set Monthly Budget	budget_settings	No
UC-5 Track Daily Limit	expenses, budget_settings	Yes (tip)
UC-6 Reminder Alerts	expenses	No
UC-7 Chat with AI	expenses, user_profile, budget_settings	Yes (chat)
UC-8 AI Suggestion	expenses, user_profile, budget_settings	Yes (suggestion)
UC-9 Monthly/Yearly Analysis	expenses, user_profile, budget_settings	Yes (analysis)
UC-10 Search/Filter	expenses	No
UC-11 Delete Expense	expenses	No